

Program	B. Tech. (SoCS/SoAE)	Semester	I
Course	Advanced Engineering Mathematics I	Course Code	MATH1059
Session	July-Dec 2025	Unit II	Integral Calculus

1. Evaluate the following integrals:

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| (i) $\int_0^1 \int_0^{\sqrt{1-x^2}} \left(\frac{xy}{\sqrt{1-y^2}} \right) dx dy$ | (ii) $\int_1^\infty \int_{x^2}^\infty \left(\frac{1}{x^4+y^2} \right) dy dx$ |
| (iii) $\int_0^1 \int_0^y \left(\frac{2xy^5}{\sqrt{1+x^2y^2-y^4}} \right) dx dy$ | (iv) $\int_2^a \int_2^b \left(\frac{1}{xy} \right) dy dx$ |
| (v) $\int_0^1 \int_{-\sqrt{y}}^{-y^2} [xy(x+y)] dx dy$ | (vi) $\int_0^2 \int_{y/4}^{3-y} [x^2 + y^2] dx dy$ |
| (vii) $\int_0^1 \int_0^{\sqrt{1+x^2}} \left(\frac{1}{x^2+y^2+1} \right) dy dx$ | (viii) $\int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \sin(\theta + \phi) d\theta d\phi$ |
| (ix) $\int_0^{\frac{\pi}{2}} \int_0^{a(1+\sin \theta)} r^2 \cos \theta d\theta dr$ | (x) $\int_0^{\frac{\pi}{4}} \int_0^{\sqrt{\cos 2\theta}} \frac{r}{(1+r^2)^2} dr d\theta$ |

2. Change the order of integration and hence evaluate the following integrals:

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| (i) $\int_0^{1/2} \int_0^{\sqrt{1-4y^2}} \left(\frac{1+x^2}{\sqrt{1-x^2}\sqrt{1-x^2-y^2}} \right) dy dx$ | (ii) $\int_0^1 \int_x^{1/x} \frac{y}{(1+xy)^2(1+y^2)} dy dx$ |
| (iii) $\int_0^1 \int_0^{\sqrt{1-x^2}} \left(\frac{e^y}{(e^y+1)\sqrt{1-x^2-y^2}} \right) dy dx$ | (iv) $\int_0^1 \int_0^{\sqrt{1-y^2}} \left(\frac{\cos^{-1} x}{\sqrt{1-x^2}\sqrt{1-x^2-y^2}} \right) dy dx$ |

3. Evaluate the following integrals by changing them into polar coordinates:

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| (i) $\int_0^1 \int_{\sqrt{x-x^2}}^{\sqrt{1-x^2}} \left(\frac{xye^{-(x^2+y^2)}}{x^2+y^2} \right) dx dy$ | (ii) $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$ |
| (iii) $\int_0^2 \int_0^{\sqrt{2x-x^2}} \frac{x}{x^2+y^2} dy dx$ | |

4. (i) Evaluate $\int \int xy(x+y) dx dy$, over the region enclosed by the parabolas $x^2 = y$ and $y^2 = -x$.

(ii) Evaluate $\int \int \frac{y}{(a-x)\sqrt{ax-y^2}} dx dy$ over the region bounded by the parabola $y^2 = x$ and the line $y = x$.

(iii) Evaluate $\int \int (x^2 - y^2) dx dy$ over the triangle with the vertices (0, 1), (1, 1) and (1, 2).

(iv) Evaluate $\int \int \frac{1}{x^4+y^2} dx dy$, over the region bounded by $y \geq x^2$, $x \geq 1$.

(v) Evaluate $\int \int \sqrt{(4x^2 - y^2)} dx dy$ over the region bounded by the lines $y = 0$, $y = 4x$, $x = 1$ and $x = y$.

(vi) Evaluate the integral $\int \int \frac{4xy}{x^2+y^2} e^{-x^2-y^2} dx dy$, over the region bounded by the circle $x^2 + y^2 - x = 0$ in the first quadrant.

- (vii) Evaluate the integral $\int \int \frac{(x^2+y^2)^2}{x^2y^2} dx dy$, over the region common to the circles $x^2 + y^2 = ax$ and $x^2 + y^2 = by$ ($a, b > 0$).
- (viii) Evaluate $\iint r^4 \cos^3 \theta dr d\theta$, over the interior of the circle $r = 2a \cos \theta$ ($a > 0$).
5. Find the area of the curvilinear triangle bounded by the parabolas $y^2 = 12x$, $x^2 = 12y$, circle $x^2 + y^2 = 45$ and lying outside the circle.
6. Find the area of the crescent bounded by the circles $r = \sqrt{3}$ and $r = 2 \cos \theta$.
7. Find the area enclosed by the curves $y^2 = 2(2 - x)$ and $y = 2 - x$.
8. Find the area of the smaller region bounded by the circle $x^2 + y^2 = 9$ and a straight line $x = 3 - y$.
9. (i) Using the transformation $x - y = u$, $x + y = v$, evaluate $\iint \cos\left(\frac{x-y}{x+y}\right) dx dy$ over the region bounded by the lines $x = 0$, $y = 0$, $x + y = 1$.
- (ii) Evaluate $\iint_R (x + y)^2 dx dy$, where R is the parallelogram in the xy -plane with vertices $(1, 0)$, $(3, 1)$, $(2, 2)$, $(0, 1)$, using the transformations $x + y = u$, $x - 2y = v$.
10. Evaluate the following triple integrals:
- (i) $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz dx dy dz$
- (ii) $\int_0^\infty \int_0^\infty \int_0^\infty \frac{1}{(1+x^2+y^2+z^2)^2} dx dy dz$.
11. (i) Evaluate the triple integral $\iiint \frac{1}{\sqrt{1-x^2-y^2-z^2}} dx dy dz$ over the region bounded by the sphere $x^2 + y^2 + z^2 = 1$.
- (ii) Find the volume of the tetrahedron bounded by the planes $x = 0$, $y = 0$, $z = 0$ and $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.
12. (i) If $\beta(n, 2) = 1/42$ and n is a positive integer, find the value of n .
- (ii) Prove that: $(m + n) \beta(m + 1, n) = m \beta(m, n)$.
13. Using Beta and Gamma functions, evaluate the following integrals:
- (i) $\int_0^1 (1 - x^5)^{-1/5} dx$ (ii) $\int_0^\pi x \sin^7 x \cos^4 x dx$ (iii) $\int_0^{\pi/4} \cos^3(2\theta) \sin^4(4\theta) d\theta$.
14. Find the volume of the solid bounded below by the xy - plane, bounded above by the cone $z = \sqrt{x^2 + y^2}$ and laterally by the cylinder $x^2 + y^2 = 4$, using cylindrical coordinates.
15. A cylindrical hole of radius b is bored through the center of a sphere of radius a . Find the volume of the remaining solid.
